



Community-acquired pneumonia in children: Outpatient treatment

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INTRODUCTION

Community-acquired pneumonia (CAP) is defined as an acute infection of the pulmonary parenchyma in a patient who has acquired the infection in the community, as distinguished from hospital-acquired (nosocomial) pneumonia. CAP is a common and potentially serious illness with considerable morbidity.

The outpatient treatment of CAP in infants and children in resource-abundant settings will be reviewed here. Other related topics are discussed separately.

- (See ["Neonatal pneumonia"](#).)
- (See ["Pneumonia in children: Epidemiology, pathogenesis, and etiology"](#).)
- (See ["Community-acquired pneumonia in children: Clinical features and diagnosis"](#).)
- (See ["Pneumonia in children >28 days old: Inpatient treatment"](#).)

CRITERIA FOR OUTPATIENT MANAGEMENT

Most children with mild CAP (eg, low-grade fever, mild or no respiratory distress, normoxia) (table 1) can be managed in the outpatient setting [1-4]. The decision to hospitalize a child with CAP is individualized based on age, underlying medical problems, and clinical factors including

Severity of community-acquired pneumonia in infants and children	
Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
Mild or absent respiratory distress: - No to mild tachypnea for age - Mild or absent retractions - No grunting - No nasal flaring - No apnea - Mild shortness of breath	Severe respiratory distress: - RR >70 breaths/minute for infants; RR >90 breaths/minute for older children - Severe supracostal, intercostal, or subcostal retractions (<12 months) - Severe difficulty breathing (>12 months) - Grunting - Nasal flaring - Apnea - Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status

Normoxemia (oxygen saturation $\geq 90\%$ in room air)	Hypoxemia (sustained oxygen saturation $< 90\%$ in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill < 2 seconds	Capillary refill ≥ 2 seconds

Table 1 - larger image below

Indications for inpatient management of community-acquired pneumonia in children	
Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or <i>Chlamydia trachomatis</i> is suspected and the child is clinically asymptomatic without hospital
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Apnoea	SpO ₂ <95% on room air
Dehydration	Unable to maintain oral hydration or unable to eat an infant
Reluctance to accept respiratory therapy	Example: 44 to 60 breaths per minute for infants <12 months of age, 30 to 45 breaths per minute for children >12 months of age, retractions, nasal flaring, difficulty breathing, asynchrony, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and thus suggests more severe disease
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic disorders, immunological disorders
Underlying conditions that may be exacerbated by pneumonia	Example: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Example: immunocompromised host
Complications of pneumonia present	Examples: effusions/empyema, necrotising pneumonia, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Streptococcus pneumoniae</i> , group A <i>Streptococcus</i>
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

Table 2 - larger image below

Clinical and radiographic clues to the aetiology of pneumonia in children*		
Etiology	Clinical features	Radiographic features
Bacterial	- Onset of ill at days - High fever - Expiratory crackles - Consolidation - Leucocytosis	- Consolidation - Segmental consolidation - Local consolidation "Round" pneumonia
Viral	- Reluctance to cough respiratory distress - Rapid deterioration - Unsettled sleep - WBC count $<10,000/mm^3$ (if obtained) - Exacerbated wheeze (phase 2 of illness)	- Peri-vascular hyperinflation - Long streaks - Increasing subsegmental consolidation - Hyperinflation
Atypical bacterial	- Onset of ill at days (onset in children <3 years) - High fever, gradually worsening consolidation - Leucocytosis, leukopenia, pleuritic pain, hepatomegaly, pleural effusion, and hypersegmentation - Whistling - Empysematous consolidation (empysematous abscess, parapneumonic effusion, hemolytic anemia, leukopenia, pleurothoracic hydatid cyst, emphysema, parapneumonic, septic meningitis)	- Round or segmental consolidation - Pleural or pericardial effusion - Consolidation - Localized interlobular thickening - Consolidation - Pleural thickening (20%)
Fungal	- Insidious onset - Insidious onset - Processing away from pneumonia - Leucopenia - Diffuse, bilateral consolidation	- Segmental consolidation - Dispersed bronchovascular - Architectural distortion - Architectural distortion with bronchiectasis - Architectural distortion with bronchiectasis

pneumonia in children: Clinical features and diagnosis", section on 'Bacterial versus viral etiology' and "Pneumonia in children: Epidemiology, pathogenesis, and etiology", section on 'Etiologic agents' and "Pneumonia in children >28 days old: Inpatient treatment", section on 'Pretreatment evaluation'.)

	Early-stage symptoms (acute)	Advanced-stage symptoms (chronic)
Atypical pneumonia of military origin (commonly known as trench fever) (Chlamydia pneumoniae)	- Usually in infants < 3 months of age - Acute onset - Exanthema, diffuse crackles - Rhinorrhea - Paroxysmal cough pattern - Respiratory exanthema (if <i>Shigella</i> infection)	Aggravation with interstitial infiltrates
Fungal	Aggravates geographic or environmental exposure	Rebudding in later adolescence
Non-infectious Asthma	- Onset of any age - Chronic cough - Constitutional symptoms - Exposure factors	Rebudding in later adolescence

Clinical and radiographic clues to the etiology of pneumonia in children*

Table 3 - larger image below

Children with typical bacterial pneumonia tend to have an abrupt onset of symptoms, which usually include fever, chills, cough, and focal chest pain. These children typically have some degree of respiratory distress with focal lung examination findings and require treatment with antibiotics.

Children with atypical bacterial pneumonia tend to have a gradual onset of fever with headache, malaise, and/or sore throat. Initial symptoms are followed by cough (usually nonproductive) that gradually worsens and is often accompanied by wheezing and/or rales. Fever and illness severity are often milder than with typical bacterial pneumonia. These children also require treatment with antibiotics.

Children with viral pneumonia typically have a gradual onset of cough following an upper respiratory tract infection. Lung findings are diffuse, bilateral, and often include wheezing. These patients should not be treated with antibiotics; however, some viral pathogens may require antiviral treatment. The diagnosis of these pathogens is discussed separately. (See ["Seasonal influenza in children: Clinical features and diagnosis"](#) and ["COVID-19: Clinical manifestations and diagnosis in children"](#) and ["Respiratory syncytial virus infection: Clinical features and diagnosis in infants and children"](#).)

The clinical features of bacterial and viral CAP frequently overlap, making it difficult to distinguish between bacterial and viral etiologies. Up to one-half of CAP in children may be a mix of bacterial and viral infections.

Since the introduction of pneumococcal conjugate vaccines (PCVs), the epidemiology of CAP in children has shifted. In a multicenter, prospective, observational study of 2222 children <17 years of age hospitalized for CAP, approximately two-thirds had a viral etiology [6]. Nearly all cases of pneumonia in children <5 years old were caused by a virus. In children ≥5 years old, approximately 20 to 35 percent of cases were caused by an atypical bacterium.

PRESUMED BACTERIAL ETIOLOGY

Approach to empiric therapy — Children with CAP who are treated in the outpatient setting typically are treated empirically; tests to identify a bacterial etiology are not recommended for

most children who are well enough to be treated in the outpatient setting [1,2]. Evaluation of children with more severe disease is discussed separately. (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)".)

Once the decision is made to initiate antibiotic therapy, factors that influence the choice of agent include the spectrum of likely pathogens, local antimicrobial susceptibility patterns, regimen simplicity, tolerability, palatability, safety, and cost [7]. The recommendations of most guidelines are based on the antibiotic susceptibility of the most likely pathogen or pathogens rather than on evidence of the superiority of one antibiotic over another [1,2]. The clinical response to the most commonly used antimicrobials appears to be similar, regardless of etiology [8-10].

Consultation with a specialist in infectious diseases may be helpful in children with medication allergies or comorbid conditions.

Neonates — The treatment of neonatal pneumonia is discussed separately. (See "[Neonatal pneumonia](#)".)

Children aged 1 to <6 months — Bacterial pneumonia in infants <6 months old usually causes more severe symptoms (see [table 1](#)), including moderate to severe respiratory distress and/or hypoxia (oxygen saturation <90 percent in room air at sea level). Inpatient management of these patients is generally advised, as discussed separately. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

Occasionally, children aged 1 to <3 months present with milder symptoms and without fever, but bacterial infection is still suspected on the basis of focal lung findings, staccato cough, or tachypnea following an upper respiratory tract infection. Such afebrile pneumonias are most commonly caused by *Chlamydia trachomatis* [11,12]. Outpatient management may be attempted, provided that patients are not hypoxic and remain afebrile [12]. Treatment regimens for *C. trachomatis* are discussed separately. (See "[Chlamydia trachomatis infections in newborns and young infants](#)", section on 'Treatment'.)

Bordetella pertussis is a less common but more severe cause of afebrile pneumonia in infants <4 months of age. Young infants with suspected *B. pertussis*-associated pneumonia should be admitted to the hospital because they are at risk for complications, including apnea, seizures, respiratory distress, pulmonary hypertension, and death. (See "[Pertussis infection in infants and children: Clinical features and diagnosis](#)", section on 'Infants younger than 12 months' and "[Pertussis infection in infants and children: Treatment and prevention](#)", section on 'Hospitalization'.)

Children aged six months to <5 years — For most infants ≥6 months old and children <5 years of age with presumed bacterial CAP, we suggest treatment



with high-dose [amoxicillin](#) (90 mg/kg per day; maximum dose 4 g per day) given in two to three divided doses ([table 4](#)) [1,2,13].

[Amoxicillin](#) is preferred because it provides coverage for *Streptococcus pneumoniae*, the most common bacterial etiology of CAP in this age group ([figure 1](#)), and it is well tolerated and inexpensive [1,2,4]. Of note, the detection rate of *Mycoplasma pneumoniae* in two- to four-year-olds with pneumonia is increasing [6,14] but still remains highest among school-age children and adolescents [15]. For children with clinical features strongly suggestive of atypical bacterial pneumonia, it is reasonable to treat with a macrolide instead of amoxicillin. (See '[Bacterial versus viral etiology](#)' above and '[Children aged ≥5 years](#)' below.)

Initial oral empiric antibiotics for outpatient treatment of pediatric community-acquired pneumonia

Table 4 - larger image below

Standard [amoxicillin](#) dosing (35-50 mg/kg per day; maximum dose 4 g per day) administered three times daily could be considered in fully immunized children residing in locales where the prevalence of penicillin-nonsusceptible *S. pneumoniae* is <10 percent [16,17]. Although we use high-dose amoxicillin, limited clinical trial data suggest that rates of treatment failure may be similar in young children receiving high versus standard amoxicillin dosing for outpatient treatment of CAP [18]. In a multicenter trial (CAP-IT), 814 young children (age >6 months and weight ≤24 kg) diagnosed with CAP who were initially managed in the emergency department, observation unit, or inpatient ward were randomly assigned to receive a standard dose (35 to 50 mg/kg per day) versus a high dose (70 to 90 mg/kg per day) of amoxicillin upon discharge [18]. Both groups had similar rates of treatment failure (12.6 versus 12.4 percent; absolute difference 0.2 percent). Among children with more severe illness, the between-group difference was 3.8 percent (17.3 versus 13.5 percent), but the finding was not statistically significant.

Despite these findings, we continue to use high-dose [amoxicillin](#) for patients with CAP. It is likely that many of the children enrolled in CAP-IT had viral rather than bacterial causes for their respiratory symptoms. This may explain the apparent lack of difference between high-versus standard-dose amoxicillin, particularly in milder cases. (See '[Resistance of *Streptococcus pneumoniae* to beta-lactam antibiotics](#)', section on '[Mechanism of action and resistance](#)'.)

Children aged ≥5 years — For most children ≥5 years old with presumed bacterial CAP, we suggest treatment with a macrolide antibiotic (typically with [azithromycin](#)) due to its fewer side effects compared with [erythromycin](#) and its convenient dosing schedule (azithromycin 10 mg/kg on day 1 followed by 5 mg/kg daily for four more days; maximum doses: 500 mg on day 1 and 250 mg thereafter) ([table 4](#)) [8,19,20]. However, in children whose presentation

strongly suggests a typical bacterial pneumonia (eg, abrupt onset of fever, chills, cough, and focal chest pain), [amoxicillin](#) is preferred, using the same dosing as for children <5 years of age ([table 4](#)). Combination therapy with a macrolide plus amoxicillin or monotherapy with [levofloxacin](#) ([table 4](#)) is a reasonable alternative option for children whose presentation does not clearly suggest one type of bacterial etiology over the other. (See '[Bacterial versus viral etiology](#)' above and '[Children aged six months to <5 years](#)' above.)

Macrolides provide coverage for *M. pneumoniae*, the most common cause of bacterial CAP among otherwise healthy children five years and older ([figure 1](#)) [6,14]. However, the prevalence of macrolide-resistant *M. pneumoniae* is increasing in some geographic regions, including the United States, Asia, Europe, and Israel [21-29]. The reported prevalence of macrolide resistance among *M. pneumoniae* isolates ranges from approximately <10 percent in the United States to 90 percent in China and some parts of Japan [15,23,26,30-32]. If patients treated with a macrolide are not improving as expected, a change in antibiotics may be required. (See '[Monitoring and expected course](#)' below and '[Treatment failure](#)' below.)

We favor [amoxicillin](#) over a macrolide for patients with presentations strongly suggestive of typical pneumonia because amoxicillin provides better coverage for *S. pneumoniae*, the second most common cause of bacterial pneumonia in this age group [1,6]. Additionally, using amoxicillin when *M. pneumoniae* is less likely decreases macrolide overuse, which is the driver of macrolide-resistant strains.

Special populations

Penicillin allergy — Alternative medications may be appropriate in selected children who have had reactions to penicillin antibiotics. The antibiotic choice is individualized according to the drug allergy history ([table 5](#)) and the ability to safely conduct an oral challenge if necessary [33,34]. This is discussed in detail separately. (See "[Choice of antibiotics in penicillin-allergic hospitalized patients](#)" and "[Penicillin allergy: Immediate reactions](#)" and "[Penicillin allergy: Delayed hypersensitivity reactions](#)".)

For children with mild reactions to penicillin and without features of an immunoglobulin E (IgE)-mediated or serious delayed reaction ([table 5](#)), [amoxicillin](#) is likely safe. An extended-spectrum cephalosporin (eg, [cefuroxime](#) axetil, [cefpodoxime](#)) is an acceptable alternative in communities with low rates of penicillin-resistant *S. pneumoniae* isolates [1]. Doses are provided in the table ([table 4](#)).

For children with IgE-mediated reactions or serious delayed reactions to penicillin, options include [clindamycin](#), [levofloxacin](#), or [linezolid](#) [1,2]. Doses are provided in the table ([table 4](#)).

Inability to tolerate oral medications — For the infant or child who is suspected to have bacterial CAP and is unable to tolerate liquids at the time of presentation, a **single** initial

dose of [ceftriaxone](#) (50 to 75 mg/kg) may be administered intramuscularly (IM) or intravenously (IV) before starting oral antibiotics [35,36]. Most children will improve sufficiently after a single dose of ceftriaxone to be able to tolerate oral medications. If there is ongoing poor oral intake, hospitalization may be warranted. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)", section on 'Indications for hospitalization' and "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

Use of IM or IV [ceftriaxone](#) is generally **not** warranted in children with uncomplicated CAP who are able to tolerate liquids because it is expensive and provides no benefit over oral antibiotics.

Children with swallowing disorders — Aspiration pneumonia acquired in the outpatient setting most commonly occurs in children with swallowing dysfunction due to an underlying neuromuscular or airway disorder. (See "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)", section on 'Aspiration pneumonia' and "[Aspiration due to swallowing dysfunction in children](#)".)

We generally treat these patients with [amoxicillin-clavulanate](#) (40 to 50 mg/kg per day of the [amoxicillin](#) component divided into two or three doses; maximum dose 1.75 g per day) to provide coverage against the usual bacterial pathogens of CAP and the anaerobes of the oral flora.

Alternative antibiotic choices for patients with IgE-mediated or serious delayed reactions to penicillin ([table 5](#)) include [clindamycin](#) (for patients of all ages) and [moxifloxacin](#) (for adolescents), which are generally effective against anaerobes. Doses are provided in the table ([table 4](#)).

Alternative antibiotics with indications for use — The following antibiotics are not routinely used in patients with CAP but may be useful under special circumstances, as discussed below:

- Amoxicillin-clavulanate – High-dose [amoxicillin-clavulanate](#) (90 mg/kg per day of the [amoxicillin](#) component divided into two or three doses) is a reasonable alternative for unimmunized children aged six months to five years and those with chronic lung disease given the possibility of infection with *Haemophilus influenzae* and *Moxarella catarrhalis* in this age group.
- Fluoroquinolones – Fluoroquinolones (eg, [levofloxacin](#), [moxifloxacin](#)) ([table 4](#)) are active against a number of the pathogens responsible for CAP, including penicillin-susceptible and nonsusceptible *S. pneumoniae*, beta-lactamase-producing *H. influenzae* and *M. catarrhalis*, *M. pneumoniae* (both azithromycin-susceptible and nonsusceptible strains), and *C. pneumoniae* [37]. However, *S. pneumoniae* strains resistant to levofloxacin and moxifloxacin have been identified [15,38].

Although fluoroquinolones are not routine first-line agents for children, they may be reasonable if no other safe and effective alternative is available. A fluoroquinolone could be used if a patient without an obvious CAP complication fails high-dose [amoxicillin](#) and/or macrolide therapy. (See "[Fluoroquinolones](#)", [section on 'Children'](#).)

- Doxycycline – [Doxycycline](#) provides coverage for atypical pathogens, including macrolide-resistant *M. pneumoniae* [37-39]. It could be used as an alternative therapy for a patient with likely *M. pneumoniae* pneumonia and no obvious CAP complication who fails macrolide therapy. The long-held concern for enamel staining associated with doxycycline use in children younger than eight years is unfounded [40,41].

Duration of treatment — In most children six months and older who are receiving outpatient treatment for confirmed or suspected bacterial CAP and are clinically improving, we suggest five days of treatment, provided adequate follow-up is in place. If appropriate follow-up is in doubt, treatment should be extended to seven days. The approach to patients who are not improving as expected is discussed below. (See '[Monitoring and expected course](#)' below and '[Treatment failure](#)' below.)

The use of a five-day course of antibiotics to treat nonsevere CAP in the outpatient setting is supported by randomized trials and meta-analyses [18,42-46]. In one meta-analysis of four trials, 1541 children were randomized to receive a shorter course (three to five days) or a longer course (7 to 10 days) of antimicrobial therapy to treat uncomplicated CAP [45]. The rates of antibiotic failure within one month of randomization were similar between groups (7.9 versus 8.0 percent; absolute risk difference 0.1 percent, 95% CI -3.0 to 2.0 percent). Adverse events were also similar and were typically mild (eg, diarrhea, rash). [Amoxicillin](#) was the most commonly used antibiotic in the trials, followed by [amoxicillin-clavulanate](#) and [cefdinir](#).

Although these results suggest that three to five days of antimicrobial therapy may be sufficient for the outpatient treatment of uncomplicated CAP in resource-abundant settings, we continue to treat for at least five days. The mean age of study participants was <37 months, an age group in which viral pathogens predominate [45]. In the one trial that tested for respiratory viruses, viruses were detected in >80 percent of participants. Consequently, the results of the meta-analysis cannot be generalized to older children, who are more likely to have bacterial pathogens.

The available clinical data in resource-limited settings support <5 days of antibiotic therapy for outpatient treatment of pneumonia [47-49]. However, these data cannot be generalized to resource-abundant settings. In resource-limited settings, pneumonia is typically diagnosed according to World Health Organization criteria using respiratory rate thresholds and clinical findings [50]; many cases may be caused by viral pathogens.

PRESUMED OR CONFIRMED VIRAL ETIOLOGY

Supportive care is the mainstay of management for most viral pneumonia. However, even if a viral etiology is confirmed, ill-appearing patients with an abrupt onset of fever, chills, cough, and focal chest pain may also have a concurrent bacterial infection. In this case, treatment for both would be warranted. (See ['Supportive care'](#) above and ['Bacterial versus viral etiology'](#) above.)

Commonly circulating viruses ([table 6](#)) predominate in CAP presenting in early childhood and should **not** be treated with antibiotics. Antiviral agents are typically not available to treat most viral pneumonia in the outpatient setting either. Although most viral etiologies are treated with supportive care alone, exceptions include influenza and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

Antiviral treatment (eg, [oseltamivir](#)) should be initiated as soon as possible in children with suspected influenza who are at high risk for complications of influenza pneumonia ([table 7](#)). Influenza is suspected when the virus is circulating in the community and the patient has typical symptoms such as fever, headache, sore throat, rhinorrhea, cough, and diffuse myalgia. The diagnosis and management of influenza in children are discussed separately. (See ["Seasonal influenza in children: Clinical features and diagnosis"](#) and ["Seasonal influenza in children: Management"](#), section on ['Antiviral therapy'](#).)

Outpatient treatment with an antiviral agent (eg, [remdesivir](#), [nirmatrelvir-ritonavir](#)) may be indicated for select children with documented SARS-CoV-2 infection who are at high risk for progression to severe disease. The management of SARS-CoV-2 in children is discussed separately. (See ["COVID-19: Management in children"](#).)

Respiratory syncytial virus (RSV), which causes bronchiolitis in children <2 years of age, is primarily treated with supportive care. RSV may be prevented by the administration of immunoprophylaxis to infants. Immunoprophylaxis and the management of RSV bronchiolitis are discussed separately. (See ["Respiratory syncytial virus infection in infants and children: Prevention"](#) and ["Respiratory syncytial virus infection: Treatment in infants and children"](#) and ["Bronchiolitis in infants and children: Treatment, outcome, and prevention"](#).)

MONITORING AND EXPECTED COURSE

We schedule follow-up within 24 to 48 hours for children who are treated for CAP as outpatients (with or without antibiotics) [1,2]. Follow-up may be conducted in the office or performed by phone. Children with uncomplicated CAP whose symptoms are improving do not require follow-up radiographs. This includes children who received radiographs at

diagnosis and were found to have round pneumonia (ie, pulmonary consolidation that appears spherical) [51].

Children who are appropriately treated for CAP generally show signs of improvement within 48 to 72 hours and continue to gradually improve with time as follows [52]:

- Children who are recovering from typical or atypical bacterial pneumonia may continue to have a residual cough for several weeks and have moderate dyspnea on exertion for two to three months [52].
- Children who are recovering from viral pneumonia usually have a resolution of their symptoms by one month; rarely, the cough may last for up to three to four months.

Although gradual resolution of bacterial pneumonia symptoms is expected, if cough does not start to improve within two to three weeks of antibiotic completion, we evaluate for treatment failure. (See '[Treatment failure](#)' below.)

Patients whose symptoms completely resolve do not require imaging. Several studies have evaluated the utility of follow-up radiographs in cohorts of children with acute, radiologically proven CAP [53-58]. Three of the studies included clinical and radiologic follow-up at three to seven weeks after initial diagnosis [53-56]. In each of these studies, follow-up radiographs were normal or improved in asymptomatic children. Residual findings, even when present, did not result in additional therapy.

Most otherwise healthy children who develop pneumonia recover without any long-term sequelae [59]. Some prospective studies suggest that pneumonia in childhood is associated with subsequent symptoms of asthma that may persist into adulthood. However, it is not clear whether this is due to unrecognized asthma at the time of pneumonia presentation or a CAP infection predisposing children to develop asthma [60,61].

TREATMENT FAILURE

Lack of improvement — Most children with appropriate management show signs of improvement within 48 to 72 hours, with further gradual improvement in the coming weeks. When patients do not improve as anticipated, the following possibilities must be considered [1,2,62,63]:

- Alternative or coincident diagnoses (eg, foreign body aspiration) (see "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Differential diagnosis')
- Development of complications (see "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Complications')

- Ineffective antibiotic coverage (lack of coverage for the actual etiology or resistant organism) (see ["Community-acquired pneumonia in children: Clinical features and diagnosis"](#), section on 'Microbiologic testing')
- Underlying or comorbid conditions (eg, immunodeficiency, anatomic abnormality) (see ["Community-acquired pneumonia in children: Clinical features and diagnosis"](#), section on 'Differential diagnosis')

The history should be reviewed again with special attention to the possibility of foreign body aspiration and geographic or environmental exposures associated with pathogens that were not treated by the empiric regimen ([table 8](#)).

In children with persistent symptoms two to three weeks after antibiotic completion, follow-up radiographs may be helpful in assessing alternate diagnoses or coincident conditions [1,2,64]. Other conditions that should be considered if a round opacity is present include congenital lung sequestration, pulmonary arteriovenous malformation, metastatic Wilms tumor, cavitory necrosis, pleural pseudocyst, and primary lung carcinoma [51,64-68]. (See ["Community-acquired pneumonia in children: Clinical features and diagnosis"](#), section on 'Differential diagnosis'.)

In patients who fail to improve but have not worsened, we either add or strengthen coverage for *S. pneumoniae* or *M. pneumoniae* if either of these organisms was not covered with the initial therapy [1,12]. Doses are provided in the table ([table 4](#)).

- **Initial monotherapy with a beta-lactam antibiotic** – For most patients who do not improve despite treatment with a beta-lactam antibiotic ([amoxicillin](#), [amoxicillin-clavulanate](#), or cephalosporin) alone, we suggest the addition of a macrolide antibiotic to cover atypical bacteria.
- **Initial monotherapy with a macrolide antibiotic** – For most patients who do not improve despite treatment with a macrolide antibiotic alone, we suggest obtaining (if available) a throat swab for *M. pneumoniae* polymerase chain reaction (PCR) testing.

If the *M. pneumoniae* PCR is positive, the pathogen may be resistant to macrolides. In this case, the macrolide antibiotic should be changed to [levofloxacin](#) or [doxycycline](#). (See ["Mycoplasma pneumoniae infection in children"](#), section on 'Macrolide resistance'.)

For patients with a negative *M. pneumoniae* PCR, or if PCR is not available, we suggest adding high-dose [amoxicillin](#) to provide better coverage for *S. pneumoniae*. Although other options include adding high-dose [amoxicillin-clavulanate](#), an extended-spectrum cephalosporin (eg, [cefuroxime](#) axetil, [cefprozime](#)), [linezolid](#), or [clindamycin](#), we prefer high-dose amoxicillin over these options because it is well tolerated and inexpensive.

- **Treatment failure while on therapy with a beta-lactam plus a macrolide** – For patients who fail to improve while being treated with both a beta-lactam and a macrolide, we suggest changing to [clindamycin](#), [linezolid](#), or a fluoroquinolone (eg, [levofloxacin](#), [moxifloxacin](#)).

A lack of response to both a beta-lactam and a macrolide may indicate that the infection is caused by a resistant organism (eg, penicillin-nonsusceptible *S. pneumoniae*, beta-lactamase-producing *H. influenzae*, macrolide-resistant *M. pneumoniae*) or *Staphylococcus aureus*. A viral etiology is also possible. [Clindamycin](#) and [linezolid](#) provide coverage for most *S. aureus* and *S. pneumoniae* (although resistance of *S. aureus* to clindamycin should be taken into consideration), while fluoroquinolones provide coverage for penicillin-nonsusceptible *S. pneumoniae*, *H. influenzae*, and macrolide-resistant *M. pneumoniae*.

Worsening condition — Patients who develop worsening signs and symptoms (eg, moderate to severe respiratory distress, hypoxia, toxic appearance, dehydration, or inability to maintain hydration orally) generally require hospitalization. Inpatient management of CAP is discussed separately. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

These patients should also undergo additional evaluation including chest radiographs, laboratory tests, and microbiologic studies. These are discussed in greater detail separately. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)", section on 'Pretreatment evaluation'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Pediatric pneumonia](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient education" and the keyword[s] of interest.)

- Basics topics (see ["Patient education: Pneumonia in children \(The Basics\)"](#) and ["Patient education: Pneumonia in children – Discharge instructions \(The Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- **Definition** – Community-acquired pneumonia (CAP) is defined as an acute infection of the pulmonary parenchyma in a patient who has acquired the infection in the community. The clinical manifestations and diagnosis of CAP are discussed separately. (See ["Community-acquired pneumonia in children: Clinical features and diagnosis"](#).)
- **Criteria for outpatient management** – Most otherwise healthy children ≥ 6 months old with mild CAP (eg, nontoxic appearance, mild or no respiratory distress, normoxia) ([table 1](#)) can be managed in the outpatient setting. Indications for inpatient management are listed in the table ([table 2](#)).
- **Bacterial versus viral etiology** – The diagnosis of bacterial versus viral CAP is generally made clinically ([table 3](#)). (See ["Bacterial versus viral etiology"](#) above.)

A typical bacterial etiology (eg, *Streptococcus pneumoniae*) is more likely if symptoms (which usually include fever, chills, cough, and focal chest pain) start abruptly. These patients typically have some respiratory distress and focal lung examination findings.

An atypical bacterial etiology (eg, *Mycoplasma pneumoniae*) is more likely if symptom onset (eg, fever, headache, malaise, sore throat) is gradual and followed by cough (usually nonproductive) that gradually worsens and is often accompanied by wheezing and/or rales. Fever and illness severity are often milder than with typical bacterial pneumonia.

A viral etiology is more likely in children < 5 years old whose cough develops gradually following an upper respiratory tract infection. Lung findings are diffuse, bilateral, and often include wheezing.

- **Presumed bacterial etiology** – Outpatient treatment of patients with presumed bacterial CAP is typically empiric. Factors that influence the choice of agent include the likely pathogens, local antimicrobial susceptibility patterns, regimen simplicity, tolerability, palatability, safety, and cost. (See ["Approach to empiric therapy"](#) above.)
- **Neonates** – The treatment of neonatal pneumonia is discussed separately. (See ["Neonatal pneumonia"](#).)

- **Age 1 to <6 months** – Bacterial pneumonia can cause more severe symptoms in patients <6 months of age than in older children ( [table 1](#)), and inpatient management is generally advised for these patients. Inpatient management is discussed separately. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

Afebrile patients aged 1 to <3 months who are thought to have afebrile pneumonia of infancy (caused by *Chlamydia trachomatis*) can be treated in the outpatient setting if they are not hypoxic and remain afebrile. Treatment regimens for *C. trachomatis* are discussed separately. (See "[Chlamydia trachomatis infections in newborns and young infants](#)", [section on 'Treatment'](#).)

- **Age six months to <5 years** – For most patients ≥6 months and <5 years of age with CAP, we suggest high-dose [amoxicillin](#) (90 mg/kg per day given in two to three divided doses; maximum dose 4 g per day) rather than other antibiotics or standard doses of amoxicillin (35 to 50 mg/kg per day) (**Grade 2C**).

We prefer [amoxicillin](#) because it provides coverage for *S. pneumoniae*, the most common bacterial etiology of CAP in this age group, and it is well tolerated and inexpensive. A macrolide (eg, [azithromycin](#)) is a reasonable alternative option in patients with clinical features suggestive of atypical bacterial pneumonia, given that the detection rate of *M. pneumoniae* is increasing in two- to four-year-olds. (See '[Children aged six months to <5 years](#)' above.)

- **Age ≥5 years** – For most patients ≥5 years old with CAP, we suggest a macrolide antibiotic (typically, [azithromycin](#) 10 mg/kg on day 1 followed by 5 mg/kg daily for four more days; maximum doses: 500 mg on day 1 and 250 mg thereafter) rather than [amoxicillin](#) or other antibiotics (**Grade 2C**). However, in children whose presentation strongly suggests a typical bacterial pneumonia, amoxicillin is preferred ( [table 4](#)). Combination therapy with a macrolide plus amoxicillin or monotherapy with [levofloxacin](#) ( [table 4](#)) is a reasonable alternative option for children whose presentation does not clearly suggest one type of bacterial etiology over the other.

Macrolides provide coverage for *M. pneumoniae*, the most common cause of bacterial CAP among otherwise healthy children five years and older ( [figure 1](#)). Patients should be monitored for treatment failure since the prevalence of macrolide-resistant *M. pneumoniae* is increasing. (See '[Monitoring and expected course](#)' above and '[Treatment failure](#)' above.)

We favor [amoxicillin](#) over a macrolide for patients with presentations strongly suggestive of typical pneumonia because amoxicillin provides better coverage for *S. pneumoniae*, the second most common cause of bacterial pneumonia in this age group. Additionally, using amoxicillin when *M. pneumoniae* is less likely to be the pathogen decreases macrolide overuse, which is the driver of macrolide-resistant strains.

Alternative regimens should be used for children with a penicillin allergy. (See ['Penicillin allergy'](#) above.)

- **Duration of antimicrobial therapy** – A five-day course of antibiotic therapy is sufficient for most patients ≥ 6 months old with CAP, provided they have adequate follow-up. If appropriate follow-up is in doubt, treatment should be extended to seven days. (See ['Duration of treatment'](#) above.)
- **Monitoring clinical response** – Children who are treated for CAP as outpatients should have a follow-up visit within 24 to 48 hours. Children who are appropriately treated for CAP generally show signs of improvement within 48 to 72 hours and continue to gradually improve with time. (See ['Monitoring and expected course'](#) above.)

Children with uncomplicated CAP whose symptoms are improving do not require follow-up radiographs.

Most otherwise healthy children who develop pneumonia recover without any long-term sequelae.

- **Treatment failure** – Patients whose symptoms of CAP have not improved at follow-up should be reevaluated for potential complications. If there are no complications and symptoms have not worsened, adjustment of the antibiotic regimen to strengthen or broaden coverage for *S. pneumoniae* or *M. pneumoniae* is appropriate. (See ['Treatment failure'](#) above.)

Patients whose symptoms have worsened at follow-up generally require hospitalization.

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Topic 5987 Version 65.0

GRAPHICS

Table 1: Severity of community-acquired pneumonia in infants and children

Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
Mild or absent respiratory distress: ▪ No to mild tachypnea for age[*] ▪ Mild or absent retractions ▪ No grunting ▪ No nasal flaring ▪ No apnea ▪ Mild shortness of breath	Severe respiratory distress: ▪ RR >70 breaths/minute for infants; RR >50 breaths/minute for older children ▪ Severe suprasternal, intercostal, or subcostal retractions (<12 months) ▪ Severe difficulty breathing (≥12 months) ▪ Grunting ▪ Nasal flaring ▪ Apnea ▪ Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status
Normoxemia (oxygen saturation ≥90% in room air)	Hypoxemia (sustained oxygen saturation <90% in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill <2 seconds	Capillary refill ≥2 seconds

RR: respiratory rate.

* World Health Organization definitions of tachypnea according to age are as follows: <2 months: >60 breaths/minute; 2 to 12 months: >50 breaths/minute; 1 to 5 years: >40 breaths/minute; >5 years: >20 breaths/minute.^[1]

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1. World Health Organization. *The management of acute respiratory infections in children: Practical guidelines for outpatient care*. World Health Organization 1995.

Adapted from:

- Bradley JS, Byington CL, Shah SS, et al. *The management of community-acquired pneumonia in infants and children older than 3 months of age: Clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America*. *Clin Infect Dis* 2011; 53:e25.
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Graphic 72015 Version 8.0

Table 2: Indications for inpatient management of community-acquired pneumonia in children

Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or <i>Chlamydia trachomatis</i> is suspected and the child is relatively asymptomatic without hypoxia
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Hypoxia	SpO ₂ <90% on room air
Dehydration	Inability to maintain oral hydration or inability to feed an infant
Moderate to severe respiratory distress	Examples: RR >70 breaths per minute for infants <12 months of age, RR >50 breaths per minute for children ≥12 months of age, retractions, nasal flaring, difficulty breathing, apnea, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and may suggest more severe illness
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic syndromes, neurocognitive disorders
Underlying conditions that may be exacerbated by pneumonia	Example: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Example: immunocompromised host
Complications of pneumonia present	Examples: effusion/empyema, necrotizing process, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Staphylococcus aureus</i> , group A <i>Streptococcus</i>
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

We recommend that children with any of the above indications be hospitalized for management of CAP.

CAP: community-acquired pneumonia; RR: respiratory rate; SpO₂: peripheral capillary oxygen saturation.

Table 3: Clinical and radiographic clues to the etiology of pneumonia in children

Etiology	Clinical features	Radiographic features
Bacterial (most commonly <i>Streptococcus pneumoniae</i>)	- Children of all ages - Abrupt onset - Ill appearance - Chills - Moderate to severe respiratory distress - Focal auscultatory findings - Localized chest pain - WBC count >15,000/mL (if obtained) - Elevated acute phase reactants (if obtained)	- Alveolar infiltrates - Segmental consolidation - Lobar consolidation "Round" pneumonia Complications: - Pleural effusion/empyema - Lung abscess - Necrotizing pneumonia - Pneumatocele
Atypical bacterial (most commonly <i>Mycoplasma pneumoniae</i>)	- Children of all ages (most common in children >5 years) - Abrupt onset with gradually worsening constitutional findings (malaise, myalgia, headache, rash, conjunctivitis, photophobia, sore throat) and nonproductive cough - Wheezing - Extrapulmonary manifestations or complications (eg, polymorphous mucocutaneous eruptions, hemolytic anemia, hepatitis, pancreatitis, myopericarditis, aseptic meningitis)	- Lobar or segmental consolidation (37%) - Perihilar or peribronchial infiltrates (27%) - Localized reticulonodular infiltrates (21%) - Patchy infiltrates (15%)
Viral	- Usually children <5 years - Gradual onset - Preceding upper airway symptoms - Nontoxic appearance - Diffuse, bilateral auscultatory findings - Wheezing - May have associated rash (eg, measles, varicella)	- Interstitial infiltrates - Associated bronchiolitis: - Patchy atelectasis - Peribronchial infiltrations with air bronchograms - Hyperinflation with flattening of the diaphragms

Afebrile pneumonia of infancy (most commonly <i>Chlamydia trachomatis</i>)	▪ Usually in infants 1 to 3 months of age ▪ Insidious onset ▪ Tachypnea, diffuse crackles ▪ Rhinorrhea ▪ Staccato cough pattern ▪ Peripheral eosinophilia (if CBC obtained)	▪ Hyperinflation with interstitial infiltrates
Fungal	▪ Appropriate geographic or environmental exposure	▪ Mediastinal or hilar adenopathy
<i>Mycobacterium tuberculosis</i>	▪ Children of any age ▪ Chronic cough ▪ Constitutional symptoms ▪ Exposure history	▪ Mediastinal or hilar adenopathy

CBC: complete blood count; WBC: white blood cell.

* The clinical features frequently overlap and cannot reliably distinguish between bacterial, atypical bacterial, and viral etiologies; up to one-half of community-acquired pneumonias in children may be mixed bacterial/viral infections. Chest radiography generally is not helpful in determining the potential causative agent of pneumonia. Nonetheless, these features may facilitate decisions regarding empiric therapy.

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Table 4: Initial oral empiric antibiotics for outpatient treatment of pediatric community-acquired pneumonia

Age group	Empiric regimen
1 to 6 months	
Bacterial (not <i>Chlamydia trachomatis</i>)	Infants <3 to 6 months of age with suspected bacterial pneumonia should be hospitalized
<i>C. trachomatis</i>	Refer to UpToDate topic on <i>C. trachomatis</i> infections in the newborn
6 months to 5 years	
Typical bacterial*	Preferred regimens:
	▪ Amoxicillin [¶] 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), or
	▪ Amoxicillin-clavulanate 90 mg/kg per day of the amoxicillin component in 2 or 3 divided doses (MAX 4 g/day amoxicillin component)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction^Δ:
	▪ Amoxicillin 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), or
	▪ Amoxicillin-clavulanate 90 mg/kg per day of the amoxicillin component in 2 or 3 divided doses (MAX 4 g/day amoxicillin component), or
	▪ An extended-spectrum cephalosporin, such as cefuroxime axetil 20 to 30 mg/kg per day in 2 divided doses (MAX 1 g/day) or cefpodoxime 10 mg/kg per day in 2 divided doses (MAX 400 mg/day) in communities with a low rate of pneumococcal resistance to penicillin
	For children with IgE-mediated or serious delayed reaction to a penicillin:
	▪ Levofloxacin [◇] 16 to 20 mg/kg per day in 2 divided doses (MAX 750 mg/day), or
	▪ Clindamycin 30 to 40 mg/kg per day in 3 or 4 divided doses (MAX 1.8 g/day), or
	▪ Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day)
	In communities with a high rate of pneumococcal resistance to penicillin:
	▪ Levofloxacin [◇] 16 to 20 mg/kg per day in 2 divided doses (MAX 750 mg/day), or
	▪ Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day)
≥5 years	

<i>Mycoplasma pneumoniae</i> or <i>Chlamydia pneumoniae</i>	Preferred regimens:
	Azithromycin[¶] 10 mg/kg on day 1 followed by 5 mg/kg daily for 4 more days (MAX 500 mg on day 1 and 250 mg thereafter), or
	Clarithromycin 15 mg/kg per day in 2 divided doses (MAX 1 g/day), or
	Erythromycin 40 to 50 mg/kg per day in 4 divided doses (MAX 2 g/day as base, 3.2 g/day as ethylsuccinate), or
	Doxycycline 4 mg/kg per day in 2 divided doses (MAX 200 mg/day), or
	Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 500 mg/day); 500 mg once daily for children ≥16 years, or
	Moxifloxacin^{◇§} 400 mg once daily (≥18 years)
Typical bacterial*	Preferred regimens:
	Amoxicillin[¶] 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction^Δ:
	Amoxicillin 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), or
	An extended-spectrum cephalosporin, such as cefuroxime axetil 20 to 30 mg/kg per day in 2 divided doses (MAX 1 g/day) or cefpodoxime 10 mg/kg per day in 2 divided doses (MAX 400 mg/day) in communities with a low rate of pneumococcal resistance to penicillin
	For children with IgE-mediated or serious delayed reaction to a penicillin:
	Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 750 mg/day); 750 mg once daily for children ≥16 years, or
	Clindamycin 30 to 40 mg/kg per day in 3 or 4 divided doses (MAX 1.8 g/day), or
	Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day) for children <12 years; 20 mg/kg per day divided in 2 doses (MAX 1.2 g/day) for children ≥12 years
	In communities with a high rate of pneumococcal resistance to penicillin:
	Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 750 mg/day); 750 mg once daily for children ≥16 years, or
	Linezolid 30 mg/kg per day divided in 3 doses (MAX 1.8 g/day) for children <12 years; 20 mg/kg per day divided in 2 doses (MAX 1.2 g/day) for children ≥12 years
Aspiration pneumonia	
Community-acquired	Preferred regimens:

	▪ Amoxicillin-clavulanate 40 to 50 mg/kg per day in 2 or 3 divided doses (MAX 1.75 g/day amoxicillin component)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction^Δ:
	▪ Amoxicillin-clavulanate 40 to 50 mg/kg per day in 2 or 3 divided doses (MAX 1.75 g/day amoxicillin component)
	For children with IgE-mediated or serious delayed reaction to amoxicillin:
	▪ Clindamycin 30 to 40 mg/kg per day divided in 3 or 4 doses (MAX 1.8 g/day)
	▪ Moxifloxacin[◇] 400 mg once daily (for ≥18 years)

IgE: immunoglobulin E; MAX: maximum.

* For the infant or child who is suspected to have bacterial community-acquired pneumonia and is unable to tolerate liquids at the time of presentation, a single initial dose of ceftriaxone (50 to 75 mg/kg) may be administered intramuscularly or intravenously before starting oral antibiotics.

¶ Preferred agent.

Δ The choice is individualized according to the drug allergy history and the ability to safely conduct an oral challenge if necessary. Refer to UpToDate content on penicillin allergy.

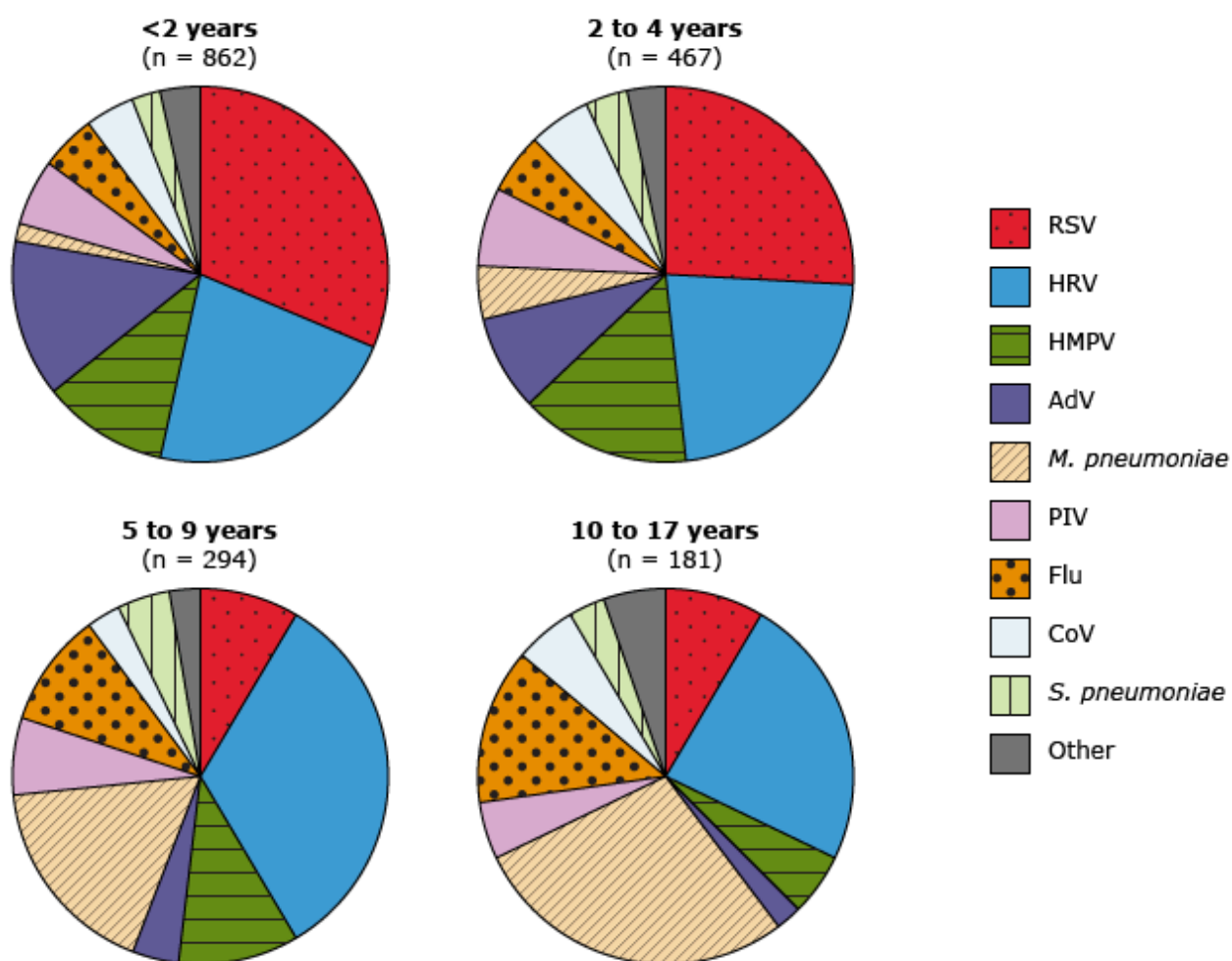
◇ In the United States, fluoroquinolones (eg, levofloxacin and moxifloxacin) are approved by the US Food and Drug Administration for community-acquired pneumonia for patients ≥18 years of age. However, they may be used in younger children if other antibiotics are inappropriate (eg, due to hypersensitivity or local antimicrobial resistance patterns).

§ Also covers typical bacterial pathogens.

Prepared with data from:

1. McIntosh K. Community-acquired pneumonia in children. *N Engl J Med* 2002; 346:429.
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Figure 1: Pathogens detected by age group in United States children with community-acquired pneumonia requiring hospitalization



AdV: adenovirus; CoV: coronavirus; flu: influenza A or B virus; HMPV: human metapneumovirus; HRV: human rhinovirus; *M. pneumoniae*: *Mycoplasma pneumoniae*; PIV: parainfluenza virus; RSV: respiratory syncytial virus; *S. pneumoniae*: *Streptococcus pneumoniae*.

From: Jain S, Williams DJ, Arnold SR, et al. Community-acquired pneumonia requiring hospitalization among U.S. children. *N Engl J Med* 2015; 372:835. Copyright © 2015 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

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Table 5: Type of drug reactions

Type of reaction	Common features
Non-allergic reaction	- Adverse effects (eg, diarrhea, vomiting, yeast vaginitis) - Family history of penicillin allergy but no personal history
Mild non-IgE-mediated reaction	- Maculopapular rash (with or without itching) - Medical record lists penicillin allergy but patient unaware of reaction
IgE-mediated reaction	- Anaphylaxis - Angioedema - Wheezing - Laryngeal edema - Hypotension - Hives/urticaria
Serious delayed reactions	- Toxic epidermal necrolysis (TEN) - Stevens-Johnson syndrome (SJS) - Drug reaction with eosinophilia and systemic symptoms/drug-induced hypersensitivity syndrome (DRESS/DiHS) - Other exfoliating dermatoses/erythroderma - Serum sickness-like reactions - Drug-induced cytopenias - Drug-induced kidney, hepatic, or other specific organ damage

IgE: immunoglobulin E.

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Table 6: Epidemiologic and clinical features of viruses that cause the common cold in children

Virus	Proportion of cases	Predominant months of circulation*	Other clinical syndromes
Rhinoviruses (more than 100 serotypes)	30 to 50%	Present year round with a peak in September and a smaller peak in March and April	
Respiratory syncytial virus	5%	November to March	■ Bronchiolitis
Influenza viruses	5 to 15%	Winter months, with peak in February	■ Influenza ■ Croup ■ Pneumonia
Parainfluenza viruses	5%	September to January with peak in October and November	■ Croup
Adenoviruses	<5%	September to May	■ Pharyngoconjunctival fever
Enteroviruses (echoviruses and coxsackieviruses)	<5%	Present year round, with a peak during the summer	■ Aseptic meningitis (nonpolio enteroviruses) ■ Herpangina (coxsackievirus type A)
Severe acute respiratory syndrome coronavirus 2	Unknown	Present year round	■ COVID-19
Common cold coronaviruses	10 to 15%	November to February	■ Pneumonia ■ Croup
Human metapneumovirus	Unknown	Late winter and early spring	■ Pneumonia ■ Bronchitis

COVID-19: coronavirus 2019.

* In the northern hemisphere.

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Graphic 56841 Version 6.0

Table 7: High-risk groups among children and adolescents for influenza complications

Category	Description
Demographic characteristics	- Children <5 years, especially those <2 years* - Children born preterm or near term¶ - Resident of a chronic care facility or nursing home
Underlying condition or treatment with common examples^Δ	
Chronic pulmonary disease	- Asthma^[1] - Cystic fibrosis - Bronchopulmonary dysplasia^[1] - Compromised respiratory function (eg, requiring mechanical ventilation, tracheostomy)
Cardiovascular disease	Hemodynamically significant conditions (excluding hypertension alone)
Kidney disease	- Chronic kidney disease, including end-stage kidney disease - Dialysis
Hepatic disease	- Chronic liver disease - Cirrhosis^[2,3]
Metabolic disorders	Diabetes mellitus
Neurologic and neurodevelopmental conditions	- Cerebral palsy - Epilepsy - Stroke - Intellectual developmental disorder - Moderate to severe developmental delay - Neuromuscular disorders, including muscular dystrophy - Spinal cord injury
Extreme obesity	BMI ≥40 for adults; BMI ≥95th percentile in children[◇]
Immunosuppression	- Receipt of immunocompromising medication - Receipt of bone marrow, hematopoietic cell transplant, and solid organ transplant - Congenital or acquired immune deficiency, including HIV - Asplenia
Receiving treatment with aspirin or salicylate-containing therapies [§]	
Pregnancy and up to 2 weeks postpartum	

BMI: body mass index; HIV: human immunodeficiency virus.

* Regardless of the presence of underlying medical conditions.

¶ Higher risk of influenza hospitalization in the first 5 years of life.

Δ List of examples is not exhaustive.

◇ BMI associated with increased risk not well defined in children but could consider BMI at or above the 95th percentile for children and teens of the same age and sex.^[4,5]

§ Applies to children and adolescents <19 years who may be at increased risk of Reye syndrome.

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Table 8: Important aspects of the history in a child with pneumonia

Historical feature	Possible significance
Age of the child	Viral etiologies are most common in infants and preschool children Atypical bacterial pathogens are more common in school-age children
Recent viral upper respiratory tract infection	May predispose to bacterial superinfection with <i>Streptococcus pneumoniae</i> or <i>Staphylococcus aureus</i>
Associated symptoms	<i>Mycoplasma pneumoniae</i> is often associated with extrapulmonary manifestations (eg, headache, sore throat, conjunctivitis, photophobia, rash)
Cough, chest pain, shortness of breath, difficulty breathing	"Classic" features of pneumonia, but nonspecific
Increased work of breathing in the absence of stridor or wheezing	Suggestive of more severe pneumonia
Fluid and nutrition intake	Difficulty or inability to feed suggests severe illness
Choking episode	May indicate foreign body aspiration
Duration of symptoms	Chronic cough (>4 weeks) suggests etiology other than acute pneumonia (refer to UpToDate topic on causes of chronic cough in children)
Previous episodes	Recurrent episodes may indicate aspiration, congenital or acquired anatomic abnormality, cystic fibrosis, immunodeficiency, asthma, missed foreign body
Immunization status	Completion of the primary series of immunizations for <i>Haemophilus influenzae</i> type b, <i>S. pneumoniae</i> , <i>Bordetella pertussis</i> , and seasonal influenza decreases, but does not eliminate, the risk of infection with these organisms
Previous antibiotic therapy	Increases the likelihood of antibiotic-resistant bacteria
Maternal history of chlamydia during pregnancy (for infants <4 months of age)	May indicate <i>Chlamydia trachomatis</i> infection
Foreign travel or known exposure to tuberculosis	May indicate <i>Mycobacterium tuberculosis</i> infection
Ill contacts	More common with viral etiologies
Travel to or residence in certain areas that suggest endemic pathogens	- Measles: Resource-limited countries - Coccidioidomycosis: Southwestern United States, northern Mexico, Central and South America - Blastomycosis: Southeastern and Central United States; states bordering the Great Lakes

	▪ Histoplasmosis: Ohio, Missouri, and Mississippi River valleys in the United States; Canada; Central America; eastern and southern Europe; parts of Africa; eastern Asia; and Australia ▪ Hantavirus: West of the Mississippi River; four corners region of United States (where borders of Colorado, New Mexico, Arizona, and Utah meet) ▪ Middle East respiratory syndrome (MERS): Countries in or near Arabian Peninsula
Animal exposure	May indicate histoplasmosis, psittacosis, Q fever
Day care center attendance	Exposure to viruses and antibiotic-resistant bacteria

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